جامعة الأمير سطام بن عبدالعزيز
PRINCE SATTAM BIN ABDULAZIZ UNIVERSITY

Attendance System Using NFC Technology

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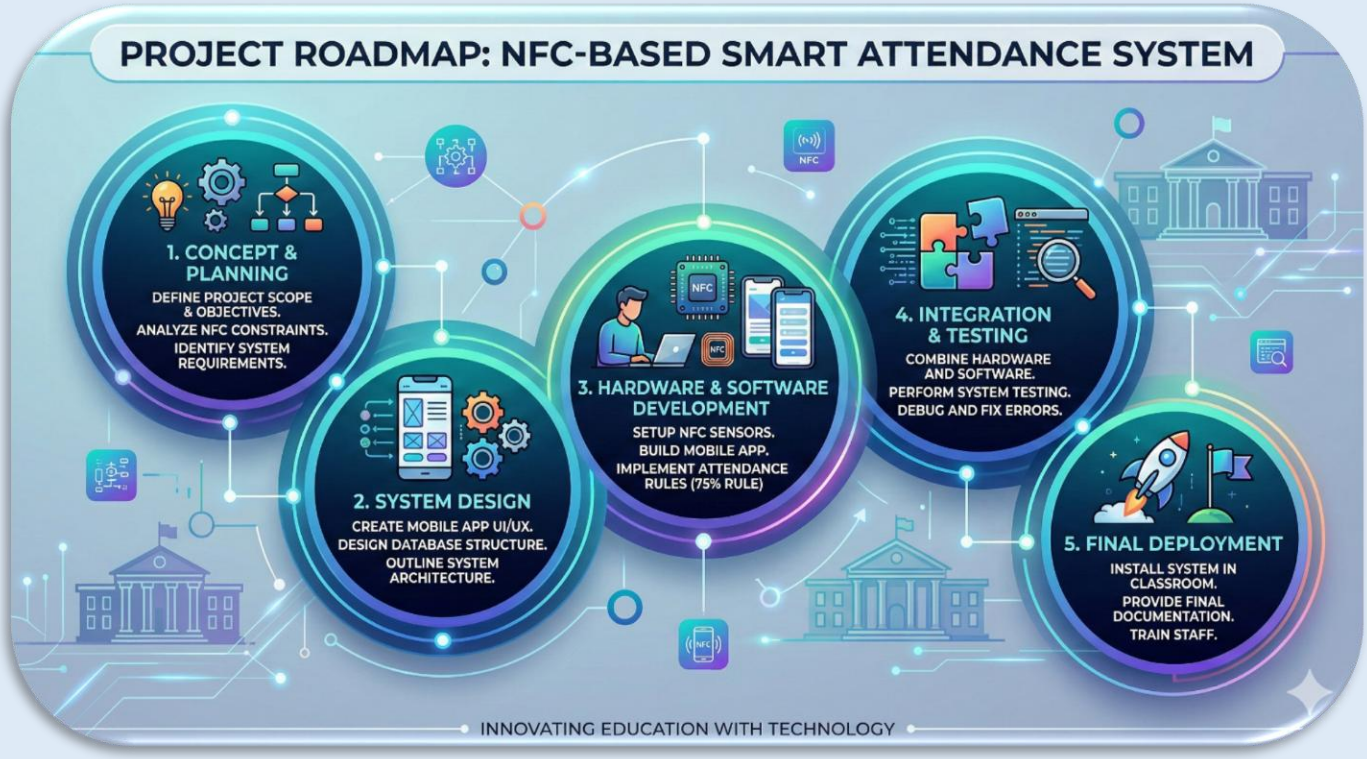
Introduction



In the rapidly evolving landscape of higher education, universities are hubs of innovation, yet many administrative tasks remain surprisingly outdated. While we teach students about the future of technology, the simple act of taking attendance often relies on archaic paper-based methods that consume valuable lecture time and are prone to manual errors. This project proposes a shift towards a "Smart Classroom" environment by introducing an Attendance System Using Near Field Communication (NFC) Technology.

The core inspiration for this project comes from the seamless "grab-and-go" technology seen in modern retail sectors, such as Amazon Fresh stores, where customers can walk out with items and payment is processed automatically without a cashier. We aim to bring this same level of "friction-free" efficiency to Prince Sattam Bin Abdulaziz University. By leveraging NFC a feature that is now standard and widely accessible in most modern smartphones we can transform the student's personal device into a secure digital ID. This proposed system is not merely about digitizing a list; it is a strategic upgrade to classroom management. It offers a "green," environment-friendly alternative by eliminating paper waste entirely and integrates directly with university systems for real-time data accuracy. Furthermore, unlike complex biometric solutions that require expensive hardware like fingerprint or face recognition scanners, this solution is highly cost-efficient, utilizing simple software and a basic NFC sensor to achieve superior results.

Problem Statement

Traditional attendance methods are becoming a burden in the modern educational environment. This project addresses several critical issues that current systems fail to solve:

- **Wasted Lecture Time:** Manual roll calls or passing around a paper sheet consumes valuable minutes that should be used for teaching

- **Incomplete Attendance:** Students often sign the attendance sheet at the beginning and leave the class early. Manual systems fail to track if a student actually stays for the required duration (75% of the lecture).

- **Human Error:** Paper-based records are prone to mistakes, lost sheets, and incorrect data entry when transferring information to the computer.

- **High Cost of Alternatives:** While digital solutions exist, technologies like fingerprint or face recognition require expensive hardware installation and complex maintenance, making them difficult to scale across all classrooms.

Background Survey

1-The "Amazon Fresh" Model

Feature	Amazon Fresh Store (Retail Model)	Proposed NFC Attendance (Education Model)
The Goal	Remove checkout lines and save shoppers' time.	Remove roll-call delays and save lecture time.
The Action	Customers take items and payment is processed automatically.	Students enter the class and attendance is recorded automatically.
The Technology	Advanced sensors detect item removal.	**NFC Sensors** detect student entry via smartphone.
The Result	A seamless, "frictionless" shopping experience.	A seamless, "frictionless" entry into the classroom.

2-NFC vs. Biometric Systems (Fingerprint)

Comparison Point	Fingerprint / Biometric Systems	Proposed NFC System
Hardware Cost	**High:** Requires installing expensive scanners in every room.	**Low:** Utilizes the student's own smartphone and a simple tag/sensor.
Setup Complexity	Complex wiring and dedicated hardware maintenance.	Simple software application and a fixed low-cost device.
Hygiene	Requires physical contact (touching the same sensor by hundreds of students).	**Contactless:** Students only touch their own personal devices.
Speed	Slow: Queues form if the scanner fails to read a print quickly.	**Instant:** NFC processing is immediate upon tapping.
Accessibility	Limited to the installed hardware capabilities.	Widely accessible as NFC is available in most modern smartphones.

Proposed Approach

Our solution is an automated attendance system consisting of a simple software application and a fixed NFC sensor installed at the classroom door.

How it works:

Registration: The student downloads the university app. To prevent fraud, each student can strictly register only **one device**. If they change phones, they must visit the IT department to update their record.

Entry: Upon entering the classroom, the student taps their phone on the NFC reader.

Verification: The system records the timestamp.

Duration Check: To solve the problem of students leaving early, the system monitors the session. A student must remain in class for at least **75% of the lecture time**. If the duration is less than 75%, the system automatically marks them as absent.

Key System Components:

- **Smartphone:** Used as the student's digital ID.
- **NFC Reader:** A low-cost sensor at the door.
- **Backend System:** Integrates with the university database.

5. Software Process Model

For the development of this system, I have selected the **Agile Process Model**.

Justification:

- **Flexibility:** If we discover that the NFC reading distance is too short or the app interface is confusing for students, Agile allows us to adapt the requirements in the next "Sprint" without scrapping the whole project.

Initial Work Plan

This timeline outlines the project execution strategy, dividing the work into clear phases to ensure timely delivery of all requirements.

Phase	Estimated Duration	Key Activities & Deliverables
1. Requirement Analysis	2 Weeks	• Define project scope and objectives. • Analyze system requirements based on NFC technology capabilities. • **Deliverable:** Project Charter & Requirements Document.
2. System Design	3 Weeks	• Design the database structure for student records. • Create User Interface (UI) mockups for the mobile app. • **Deliverable:** System Architecture & Design Prototypes.
3. Hardware Setup	1 Week	• Acquire NFC sensors and installation equipment. • Configure the reading device at the classroom door. • **Deliverable:** Hardware Environment Ready.
4. Development (Phase 1)	3 Weeks	• Develop the student mobile application. • Implement the "Single Device Registration" security logic.

Phase	Estimated Duration	Key Activities & Deliverables
5. Development (Phase 2)	3 Weeks	• Implement the "75% Attendance Rule" algorithm . • Integrate the app with the university database. • **Deliverable:** Beta Version (Full Features).
6. Testing & QA	2 Weeks	• Perform system testing to detect bugs. • Simulate student entry/exit scenarios to ensure accuracy. • **Deliverable:** Test Report & Bug Fixes.
7. Final Deployment	1 Week	• Install the system in the target environment. • Submit final documentation and user guides. • **Deliverable:** Project Handover.

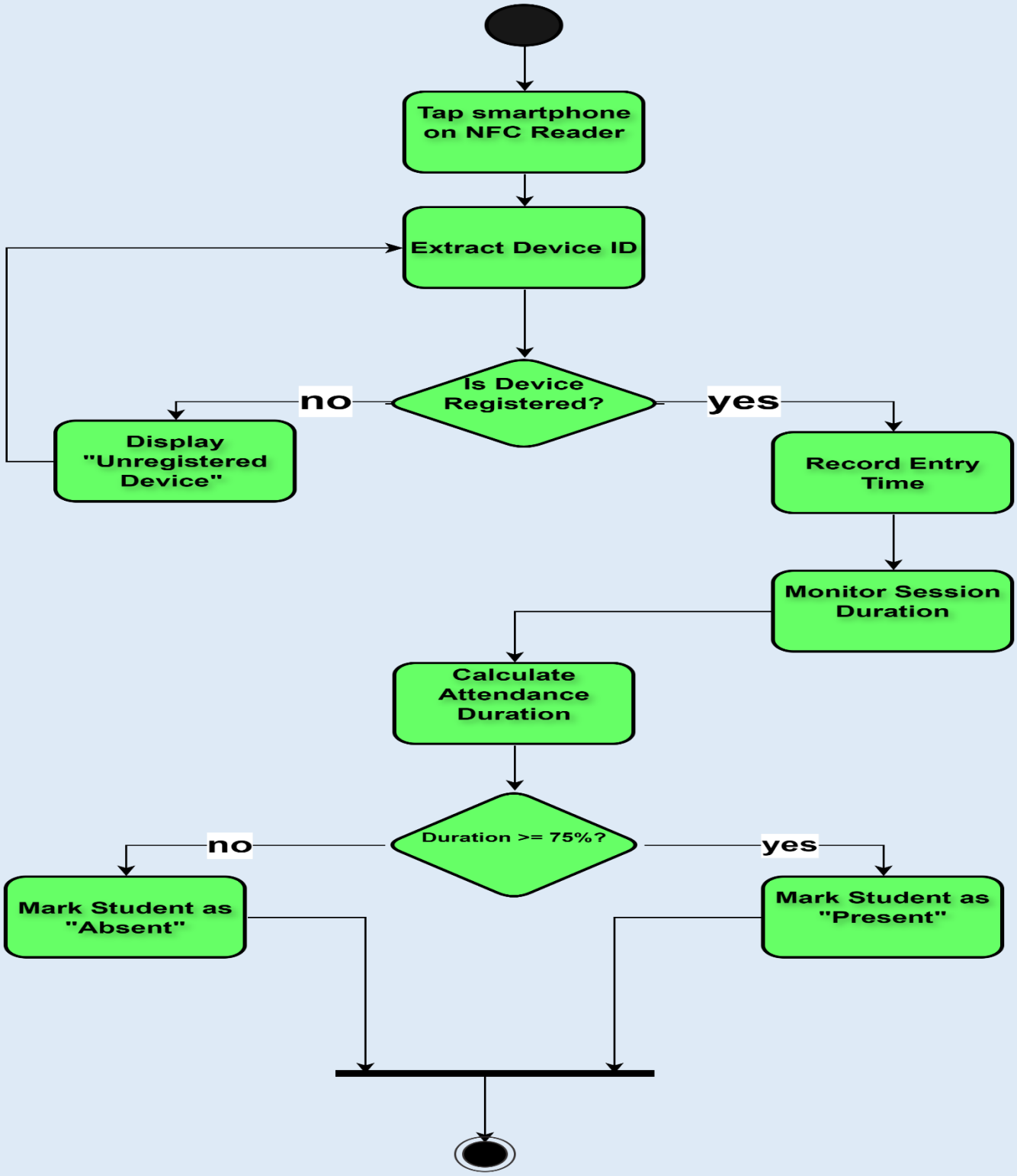
Functional Requirements

ID	Priority	Description
FR1	High	**The student** can register their personal smartphone into the system using the official university application.
FR2	High	**The student** can tap their smartphone on the classroom's NFC reader upon entering to automatically record their attendance.
FR3	Intermediate	**The system** automatically warns the student if they exceed the allowed absence percentage.
FR4	High	**The system** records the exact timestamp of the student's entry verification.
FR5	High	**The system** monitors the session duration to ensure the student remains in class for at least 75% of the lecture time.
FR6	High	**The system** automatically marks the student as absent if their recorded duration is less than 75% of the total lecture time.
FR7	High	**The system** actively prevents attendance fraud by restricting each student to strictly register only one device.
FR8	High	**The system** integrates with the backend to automatically update the central university database with the attendance records.

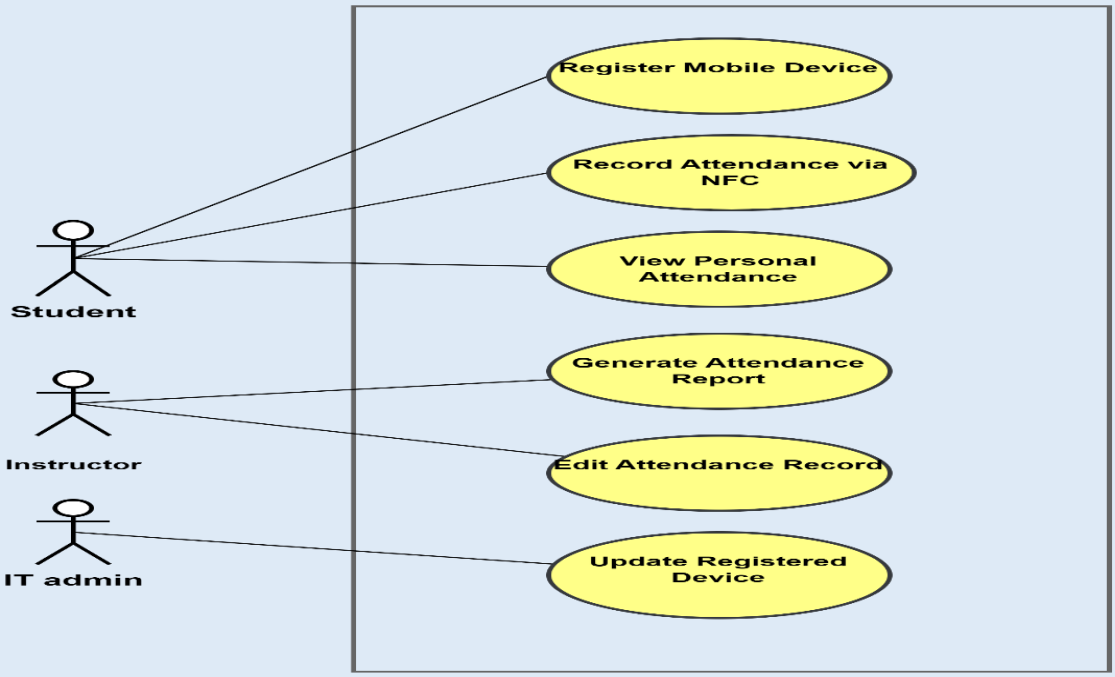
Non-Functional Requirements

ID	Priority	Description
NFR1	High	**Security:** The system must securely tie one unique smartphone to one student to prevent any form of attendance fraud.
NFR2	High	**Performance:** The system must process the NFC tap instantly upon entry to ensure a seamless experience and avoid queues.
NFR3	Intermediate	**Usability :** The system must provide a completely contactless experience where students only handle their own personal devices.
NFR4	Intermediate	**Cost-Efficiency:** The system must be able to operate using simple software and a low-cost fixed NFC sensor at the door, rather than expensive hardware.

Activity Diagram



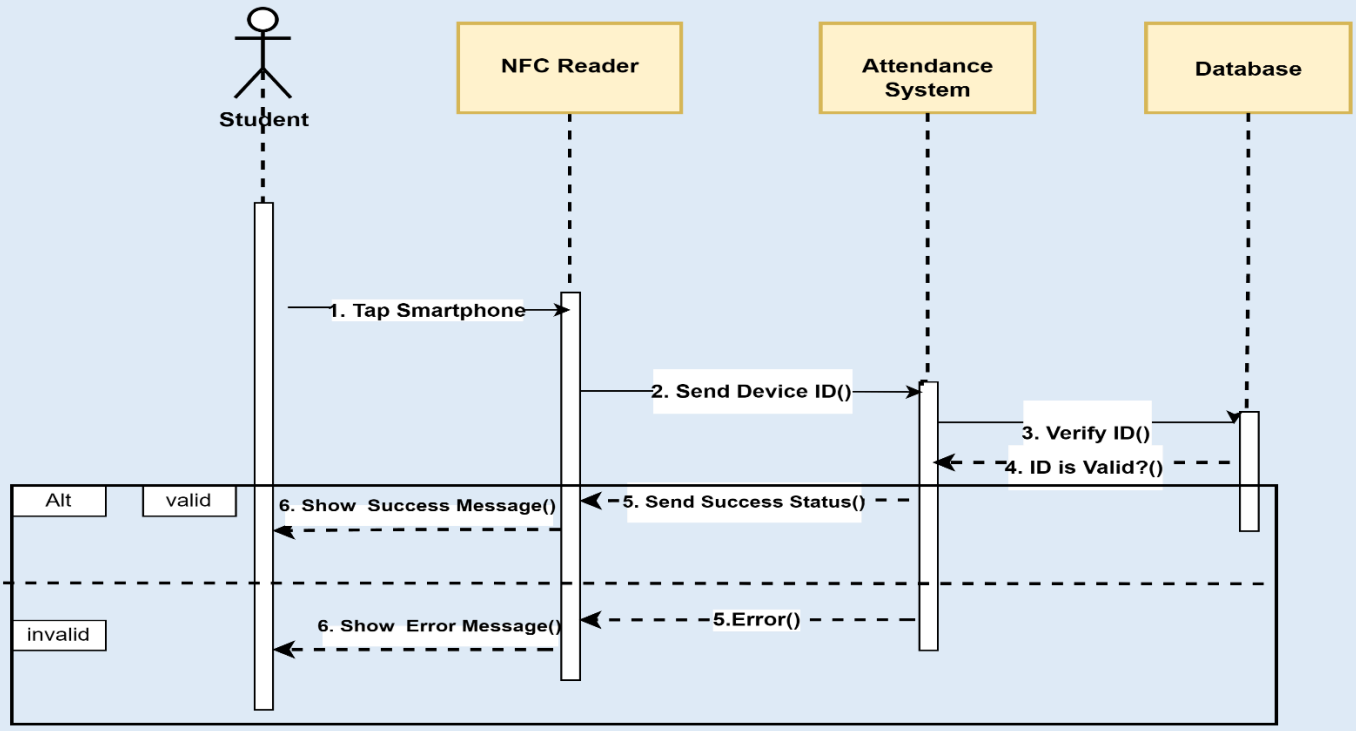
Use Case Modelling

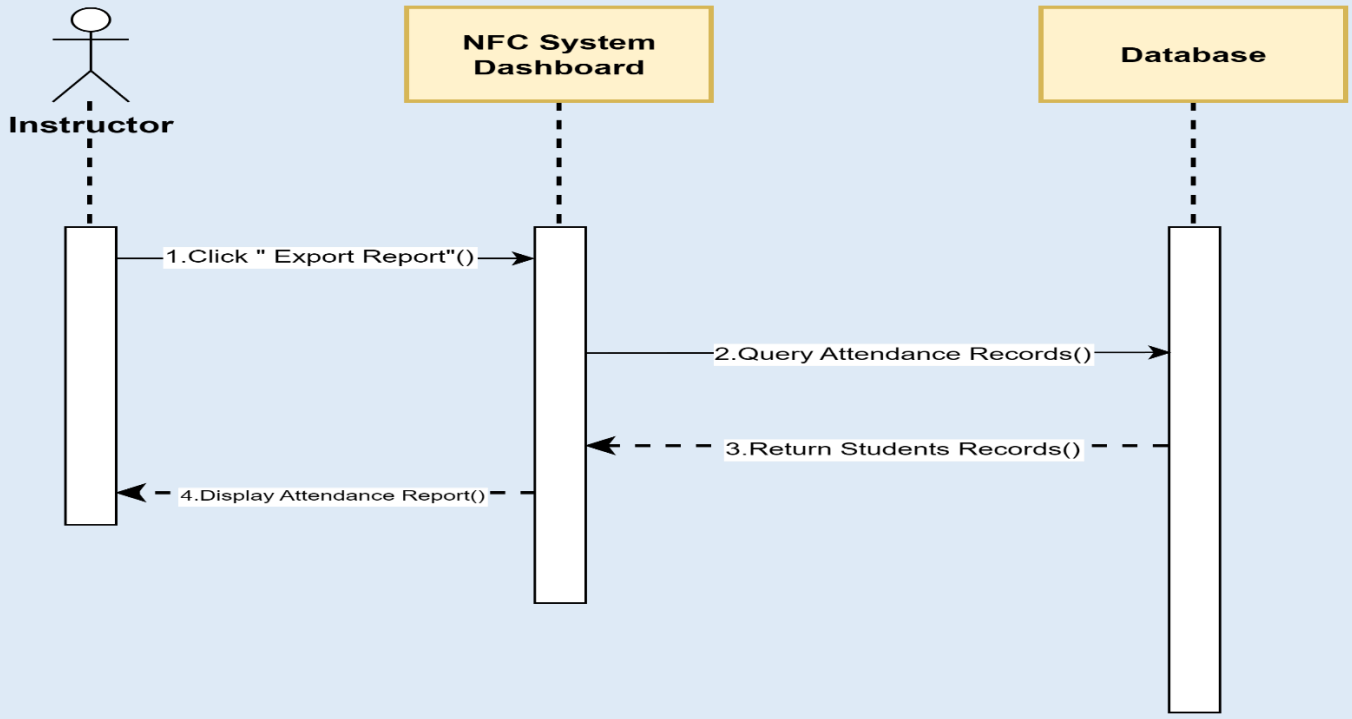


Use Case Name	**Record Attendance via NFC**
Actor	Student
Difficulty	Intermediate
Pre-condition	The student must be enrolled in the course, have the university application installed, and their smartphone must be pre-registered in the system.
Post-condition	The system successfully captures the student's entry time and initiates the session monitoring.
Processes	1. The student arrives at the classroom door. 2. The student taps their registered smartphone on the fixed NFC reader. 3. The system scans and extracts the unique Device ID. 4. The system validates the ID against the database to ensure it matches the specific student. 5. The system records the exact entry timestamp and displays a success notification.

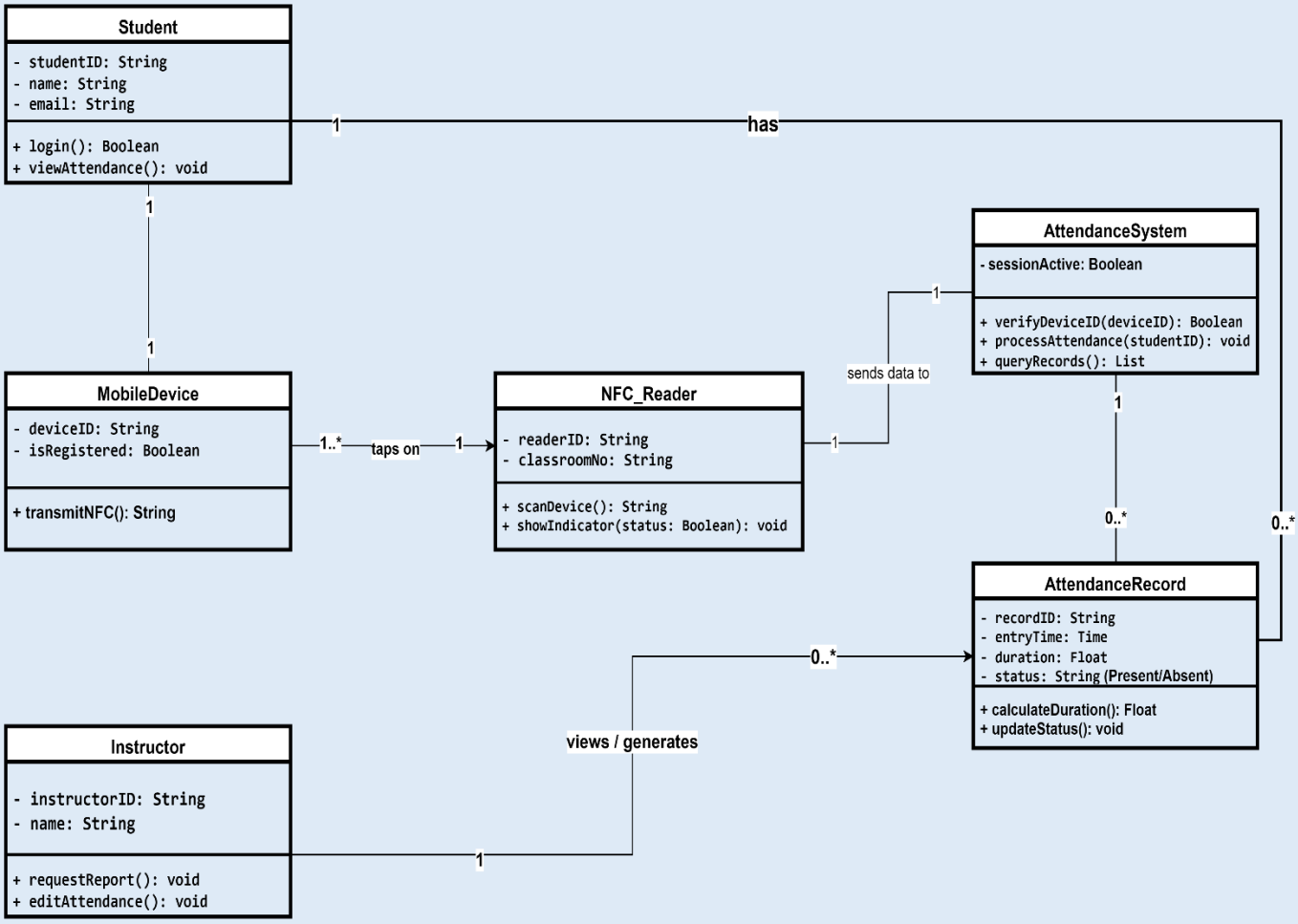
Use Case Name	**Register Mobile Device**
Actor	Student
Difficulty	Intermediate
Pre-condition	The student has downloaded the official attendance app and is logging in for the first time using their university credentials.
Post-condition	The specific smartphone is securely and permanently linked to the student's academic ID in the central database.
Processes	1. The student logs into the application. 2. The system checks if the student already has a registered device. 3. Since no device is found, the system prompts the student to link their current phone. 4. The student clicks "Confirm Registration". 5. The system securely extracts the smartphone's unique hardware identifier and saves it to the student's profile, locking the account to this specific device.

Sequence Diagrams

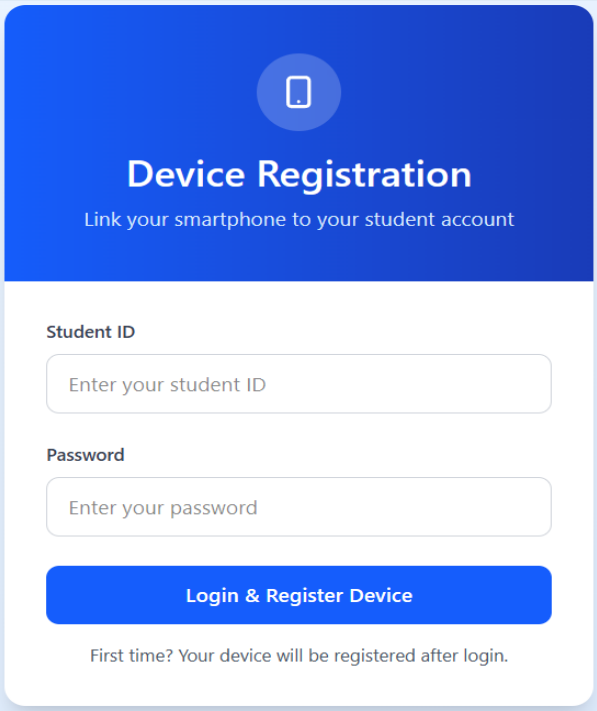


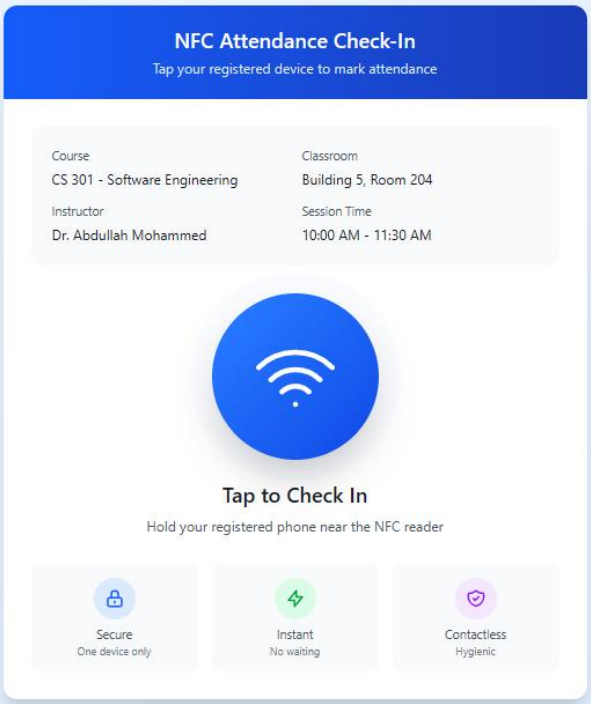


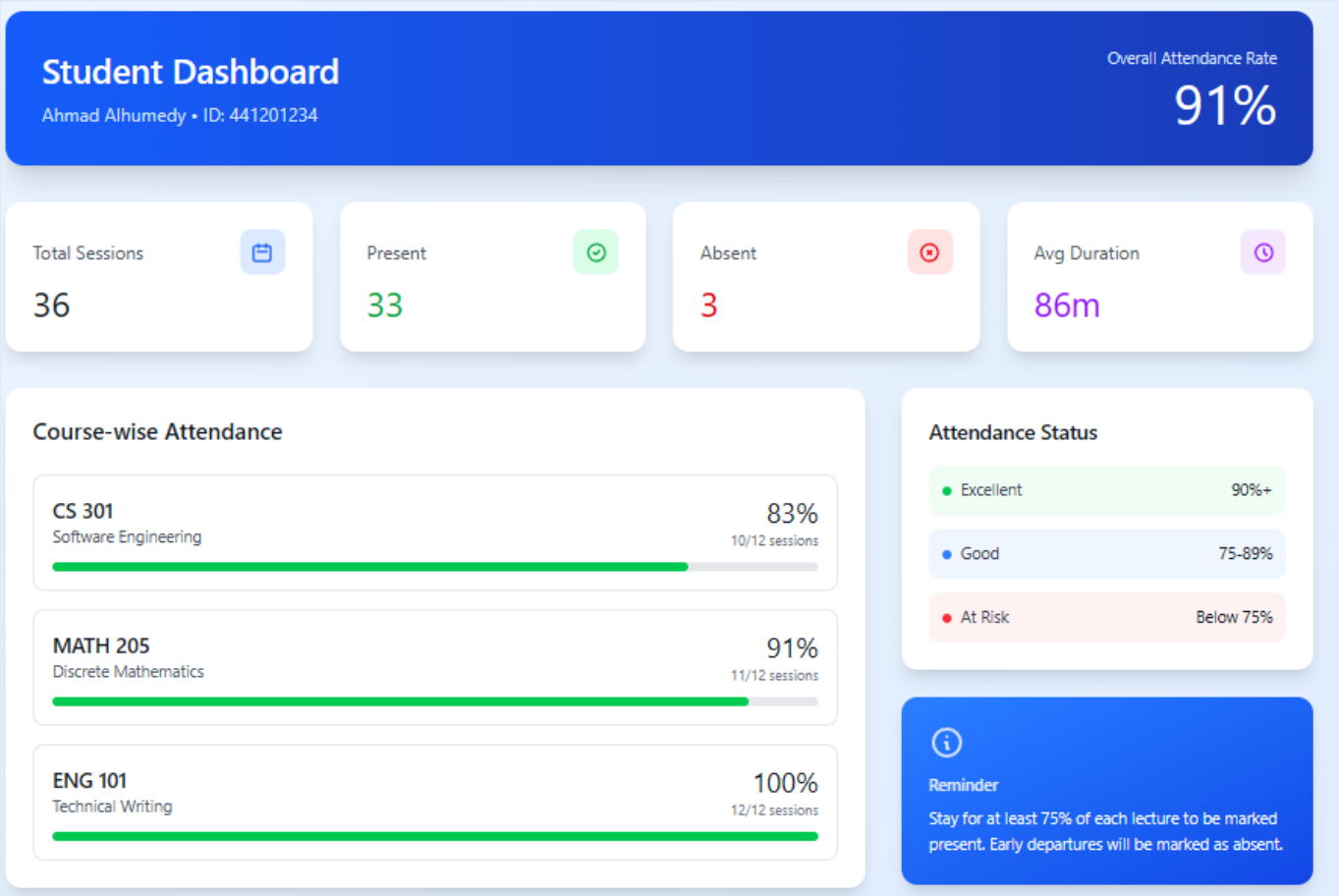
Class Diagram



User Interface Design







Instructor Dashboard

Dr. Abdullah Mohammed • College of Computer Science

Export Report

Course: CS 301 - Software Engineering | Date: 04/24/2026 | Session: 10:00 AM - 11:30 AM | Apply Filters

Total Students	Present Today	Absent Today	Avg Attendance
8	6	2	86%
Enrolled in CS 301	75% attendance rate	25% of class	Course-wide (last 30 days)

Today's Attendance - 2026-04-24

Mark Manual | Edit Records

Student ID	Name	Check-in Time	Duration	Percentage	Status	Actions
441201234	Ahmad Alhumedy	10:02 AM	85 min	94%	Present	Edit
441201235	Mohammed Ali	10:00 AM	90 min	100%	Present	Edit

Admin Panel

Device Management System • IT Department

Total Registered Devices: 5

Active Devices	Inactive Devices	Pending Updates	iOS Devices
4	1	0	3
Used in last 7 days	Not used recently	Device change requests	2 Android devices

Search by student ID, name, or email... | All Status | All Devices

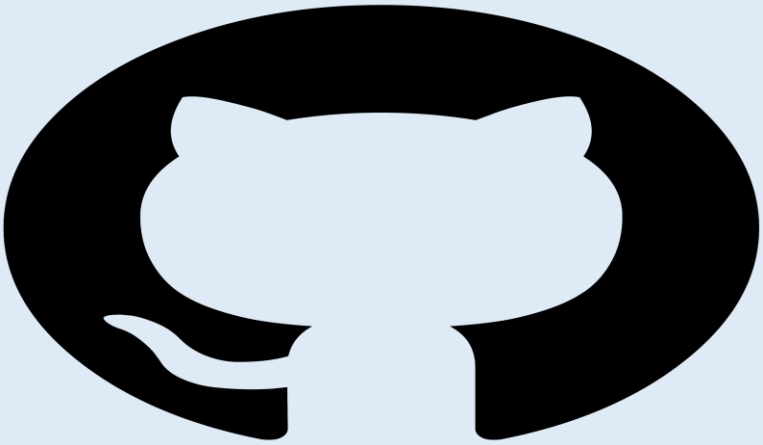
Registered Device Records

Student Info	Device Details	Device ID	Registration	Last Used	Status	Actions
Ahmad Alhumedy ID: 441201234 ahmad.h@psau.edu.sa	iPhone 14 Pro iOS 17.2	A3B7-9F2E-C4D1-8E5F	2026-01-15	2026-04-24	Active	Update Revoke
Mohammed Ali ID: 441201235 mohammed.a@psau.edu.sa	Samsung Galaxy S24 Android 14	B8C2-4F3D-9A6E-1H7G	2026-01-16	2026-04-24	Active	Update Revoke
Sarah Ahmed ID: 441201236 sarah.a@psau.edu.sa	iPhone 13 iOS 17.1	C9D3-5G4E-0B7F-2I8H	2026-01-14	2026-04-24	Active	Update Revoke

Testing

Test Case ID	Test Case Description	Input	Expected Result
T1	Verify that a student can register their smartphone for the first time.	Student logs in with valid university credentials and clicks "Register Device" using an unregistered smartphone.	The system securely extracts the hardware ID, links it permanently to the student's academic ID, and displays a "Success" message.
T2	Verify that the system prevents registering a second device to the same student account .	Student attempts to log in and register a new smartphone when an active device is already linked to their account.	The system rejects the registration, displays an "Unregistered Device" error, and instructs the student to visit the IT department.
T3	Verify successful attendance recording for a registered student arriving on time.	Student taps their previously registered smartphone on the NFC reader at the classroom door.	The reader emits a green light/success message, and the system records the exact entry timestamp.
T4	Verify system behavior when an unregistered device is tapped on the NFC reader.	Student taps a smartphone that has not been linked to any university account on the NFC reader.	The reader emits a red light/error message, and no attendance is recorded in the database.
T5	Verify that a student is marked "Present" if they stay for more than 75% of the lecture.	The system monitors a session. The student's recorded duration is calculated at 85% of the total lecture time.	The system updates the central database and marks the student's status as "Present" for that specific session.
T6	Verify that a student is marked "Absent" if they leave early and stay for less than 75% of the lecture.	The system monitors a session. The student's recorded duration is calculated at 40% of the total lecture time.	The system automatically overrides the initial check-in, updates the database, and marks the student as "Absent".

Implementation & Demo



GITHUP URL:

https://github.com/AhmadAlhumedy/NFC-Smart-Attendance-System

DEMO URL:

/https://ahmadalhumedy.github.io/NFC-Smart-Attendance-System

THE END